

Module 1: Network + Network Fundamentals and building networks

Day 3

Task 1:

Feature	IPv4	IPv6
Address length	32-bit (e.g. 192.168.1.1)	128-bit (e.g. 2001:0db8:85a3::8a2e:0370:7334)
Address format	Decimal, 4 octets separated by dots	Hexadecimal, 8 groups separated by colons
Header complexity	More complex header, includes checksum	Simplified, fixed-size header — no checksum
Configuration	Manual or DHCP	Supports auto-configuration (SLAAC) in addition to DHCPv6
Security	Security (IPSec) optional, added separately	IPSec support built into the protocol design

Task 2:

Device	IP address	Subnet mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

Task 3:

A /27 mask is 255.255.255.224, leaving 5 host bits.

- **Hosts per subnet:** $2^5 - 2 = 30$ **usable hosts**
- **Number of subnets** (carved out of the original /24): $2^3 = 8$ **subnets**

Subnet	Network	Broadcast	Usable IP
1	192.168.1.0	192.168.1.31	192.168.1.1 – 192.168.1.30
2	192.168.1.32	192.168.1.63	192.168.1.33 – 192.168.1.62
3	192.168.1.64	192.168.1.95	192.168.1.65 – 192.168.1.94

Task 4:

10.0.5.17 /24 with mask 255.255.255.0

Network address: 10.0.5.0

Broadcast address: 10.0.5.255

Valid host range: 10.0.5.1 – 10.0.5.254